

AUTONOMOUS SYSTEMS ON MARS

Part 1 - Problem Analysis

Key facts of the scenario:

- Aircraft is flying autonomously on Mars
- Dust storm detected ~12 minutes ahead on current route
- Communication with Earth has ~14-minute round-trip delay
- Aircraft must decide without real-time human control

Questions to clarify

- How accurate is the storm detection and prediction?
- How large / dense / fast-moving is the dust storm?
- What are aircraft limits (wind tolerance, visibility needs)?
- Is the aircraft solar-powered?
- What alternative routes or landing sites exist?
- How much fuel / battery reserve is available?
- What sensors are available for navigation in low visibility?
- What are mission priorities: safety vs mission completion?

Assumptions

- Aircraft has onboard autonomy and local decision capability
- Storm conditions can damage sensors or destabilize flight
- Navigation may degrade due to dust and GPS-like limitations
- Energy is limited and must be managed
- Earth operators cannot respond in time
- Communication blackout

Why the problem is hard

- Very limited decision time (12 min)

- No reliable human intervention
- Uncertain environmental data
- Safety-critical consequences
- Multiple competing objectives (survive, continue mission, conserve energy)

Part 2 — Proposed Solution

Design an onboard Autonomous Hazard Response System that evaluates risk and selects one of several actions:

1. Reroute around storm
2. Climb or descend to safer altitude
3. Enter safe-mode flight through storm
4. Abort mission and perform emergency landing
5. Hold position if storm may pass

Architecture

- Sensor fusion module (weather sensors, lidar, cameras, inertial nav)
- Environment prediction module
- Risk assessment engine
- Decision planner
- Flight control interface
- Health & energy monitoring system

Decision logic

```
if storm_risk > critical_threshold:
    if safe_landing_site_available:
        execute_emergency_landing()
    else if alternate_route_feasible:
        reroute()
    else:
        enter_storm_survival_mode()

elif storm_risk > moderate_threshold:
    adjust_altitude()
    slow_speed()
    continue_monitoring()

else:
    continue_nominal_mission()
```

Key technical choices

- Probabilistic risk model (e.g. ML predictor)
- Redundant sensors for degraded visibility
- Real-time trajectory replanning
- Energy-aware decision making
- Pre-validated emergency landing maps
- Fail-safe “survival mode” control laws

Why this approach

- Enables fully autonomous response within time limit
- Maximizes survival probability
- Maintains mission continuity when possible
- Handles uncertainty through continuous reassessment
- Modular design allows future upgrades
- Graceful degradation

